Unit: 6 Unit 1: Ratio, Rate, Percentage
Topic: Rates Variation

Name: ____________
Date: ________

Homework (Jalapeno)

Q1. In ancient Egypt, a pyramid's construction required $3 \cdot 10^3$ stone blocks per day. If the construction took 10 days, what is the total rate of blocks used?
(A) $3 \cdot 10^4$ blocks
(B) $3 \cdot 10^5$ blocks
(C) $3 \cdot 10^6$ blocks
(D) $3 \cdot 10^7$ blocks
(E) $3 \cdot 10^8$ blocks

Q2. A Roman aqueduct's water flow rate was $2 \cdot 10^2$ liters per minute. What is the flow rate in liters per hour?
(A) $1.2 \cdot 10^3$ liters/hour
(B) $1.2 \cdot 10^4$ liters/hour
(C) $2 \cdot 10^3$ liters/hour
(D) $2 \cdot 10^4$ liters/hour
(E) $2 \cdot 10^5$ liters/hour

Q3. A historian estimates that an ancient civilization's population grew at a rate of 5% per year. If the initial population was 10^3, what is the population after 2 years?
(A) $1.1025 \cdot 10^3$
(B) $1.105 \cdot 10^3$
(C) $1.1 \cdot 10^3$
(D) $1.2 \cdot 10^3$
(E) $1.5 \cdot 10^3$

Q4. An ancient trade route's length was $5 \cdot 10^2$ miles. If a merchant traveled $2 \cdot 10^1$ miles per day, how many days did the journey take?
(A) 25 days
(B) 50 days
(C) 10 days
(D) 20 days
(E) 15 days

Q5. A pharaoh's tomb required $2 \cdot 10^2$ laborers to build. If the laborers worked 8 hours a day, and the tomb took 10 days to build, what is the total labor hours?
(A) $1.6 \cdot 10^3$ hours
(B) $1.6 \cdot 10^4$ hours
(C) $3.2 \cdot 10^3$ hours
(D) $3.2 \cdot 10^4$ hours
(E) $6.4 \cdot 10^3$ hours

Q6. An ancient city's water supply system had a flow rate of $1.5 \cdot 10^2$ liters per second. What is the flow rate in liters per minute?
(A) 90 liters/minute
(B) 900 liters/minute
(C) 90 liters/minute
(D) 9000 liters/minute
(E) 90000 liters/minute

Q7. A historical artifact's value increased at a rate of 8% per year. If the initial value was

$$10^3$$

, what is the value after 3 years?
(A)
$$1.2597 \cdot 10^3$$
(B)
$$1.26 \cdot 10^3$$
(C)
$$1.2 \cdot 10^3$$
(D)
$$1.5 \cdot 10^3$$
(E)
$$2 \cdot 10^3$$

Q8. An ancient astronomical clock's gear ratio was 3 : 5. If the clock's minute hand moved $2 \cdot 10^1$ degrees, what distance did the hour hand move?
(A) 12 degrees
(B) 15 degrees
(C) 20 degrees
(D) 24 degrees
(E) 30 degrees

Open Ended Questions (FRQ)

Q9. A group of archaeologists discovered an ancient water wheel with a diameter of 5 meters. If the wheel rotated at a rate of 2 revolutions per minute, what is the linear speed of the wheel in meters per minute? Show your work and explain your reasoning.

Q10. An ancient temple's height was $2 \cdot 10^1$ meters. If a stone fell from the top of the temple and hit the ground in 2 seconds, what was the stone's average speed in meters per second? Assume a constant acceleration due to gravity of $9.8\,\mathrm{m/s^2}$. Show your work and explain your reasoning.

Chef's Tips

- Ensure students show all work and explain their reasoning for free-response questions.
- Remind students to use the correct units and significant figures.
- Encourage students to use historical and ancient contexts to inform their problem-solving approaches.